Lovely Professional University, Punjab

PROJECT REPORT

Shopping Behaviour DASHBOARD ANALYSIS

SUBMITTED TO: Pardeep Kumar

SUBMITTED BY: Mahammad Yaseen

REG. NUMBER: 12317975

SECTION : K23DF

ROLL NUMBER: 67

COURSE TITLE: DATA ANALYTICS WITH POWER BI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Introduction

In today's competitive and data-driven business environment, organizations generate and process vast amounts of customer and transactional data on a daily basis. Understanding customer behavior, identifying purchasing patterns, and monitoring key performance indicators are essential for improving business strategies and enhancing customer satisfaction. However, extracting meaningful insights from raw data becomes challenging without effective analytical tools and intuitive visual representations.

This project focuses on the design and development of an **interactive Power BI dashboard** to analyze customer shopping behavior and uncover actionable insights for business stakeholders. The dashboard integrates key information such as customer demographics, purchase frequency, product categories, spending patterns, discounts, and satisfaction metrics into a single, unified analytical view.

The dashboard is structured into two pages:

- **Page 1 – Executive Overview**, which provides high-level KPIs and summary visualizations for quick assessment of overall customer behavior and sales performance.
- **Page 2 – Detailed Analysis**, which enables deeper exploration of purchasing trends, customer segments, category-wise performance, and spending behavior using advanced Power BI visuals.

By leveraging **DAX calculations**, **advanced visualizations** (such as Decomposition Tree, Heatmaps, Box Plots, and Waterfall Charts), and **interactive slicers**, this dashboard empowers decision-makers to identify key trends, optimize marketing strategies, improve customer engagement, and drive data-driven business decisions. This project demonstrates how business intelligence tools like Power BI can transform complex customer data into clear, meaningful insights that support strategic planning and operational excellence.

LINKEDIN URL:

https://www.linkedin.com/posts/mahammad-yaseen07_powerbi-dataanalytics-dashboarddesign-activity-7405897460105084928-y6K0?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAEk3FXwBrgtY5m1MeFMyw5tHnQdInACHMVw

DATA SOURCE URL:

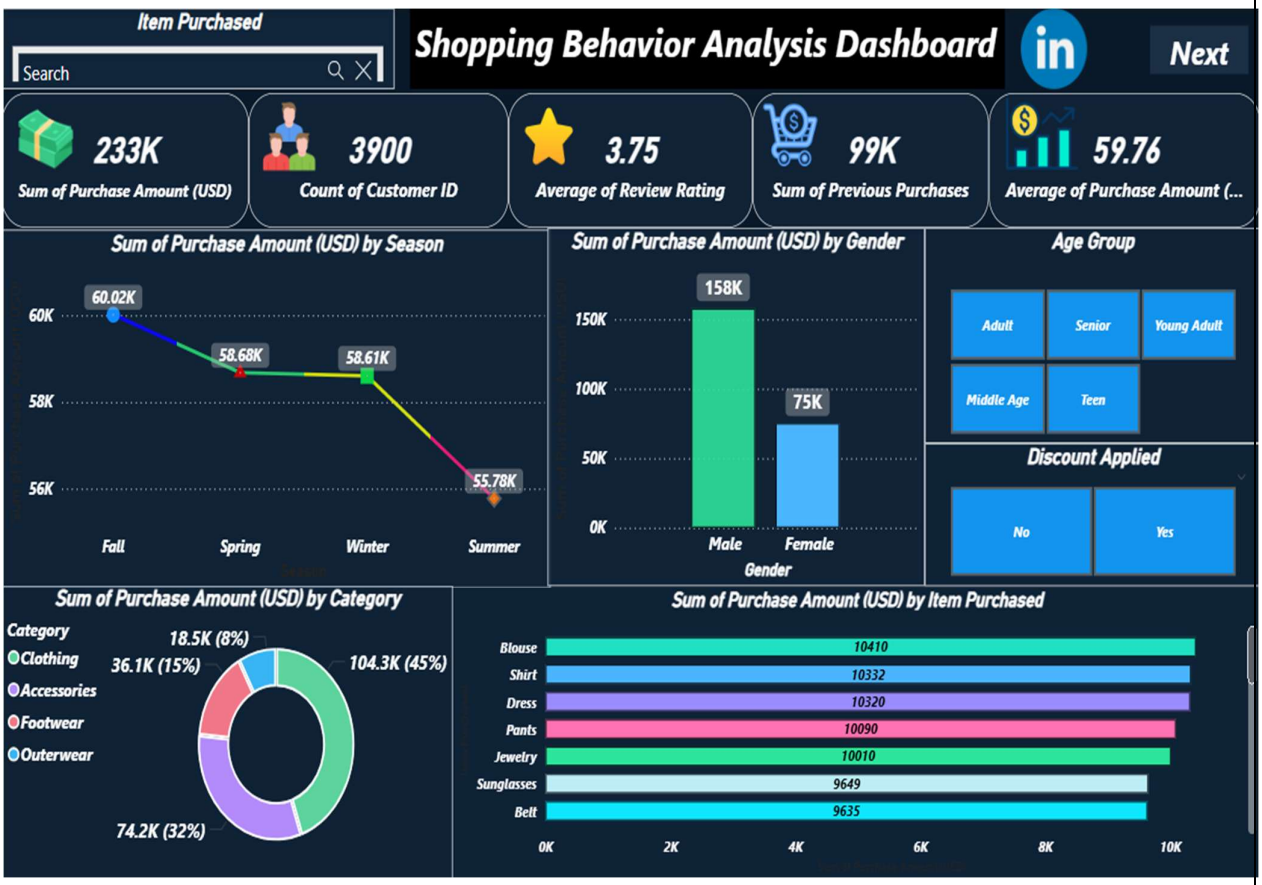
https://www.kaggle.com/datasets/saadaliyaseen/shopping-behaviour-dataset

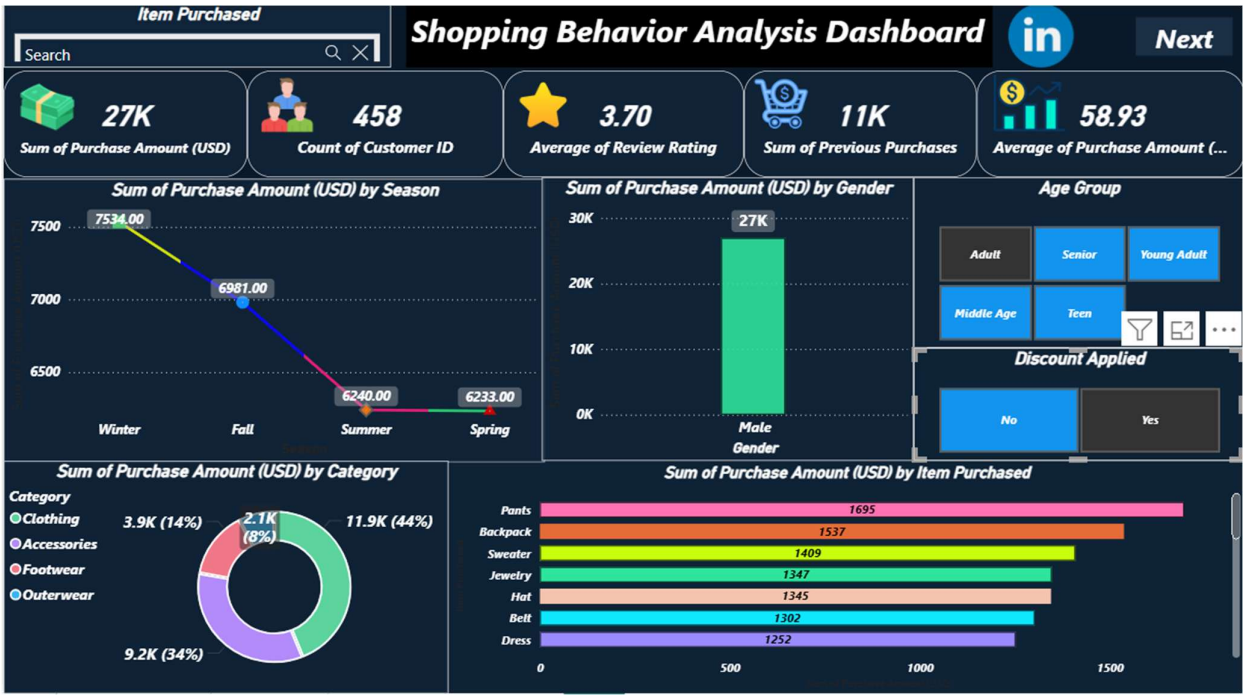
GOOGLE DRIVE URL:

https://drive.google.com/drive/folders/1sWQRbEJuhOkrbgN3e5MglA1_rEWfRiBd?usp=drive_link

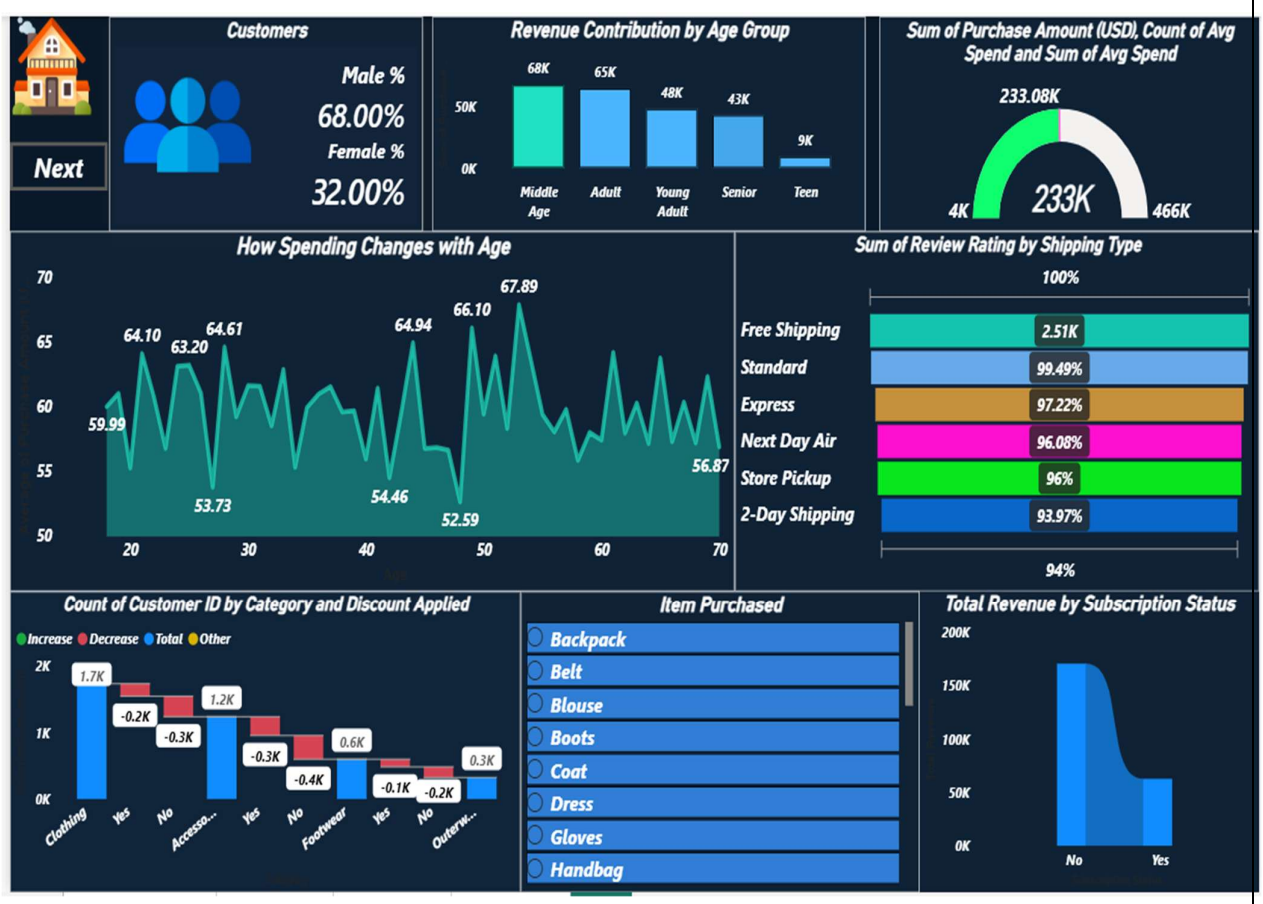
SCREENSHOTS:

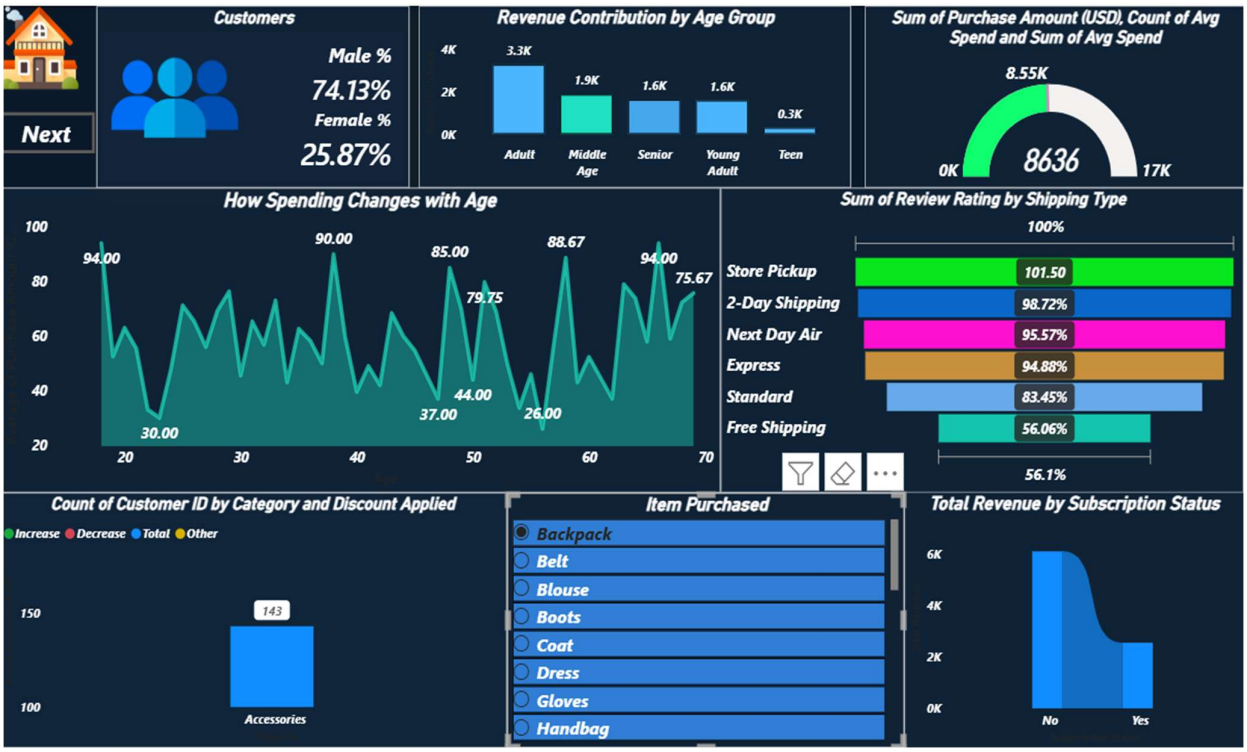
PAGE 1-



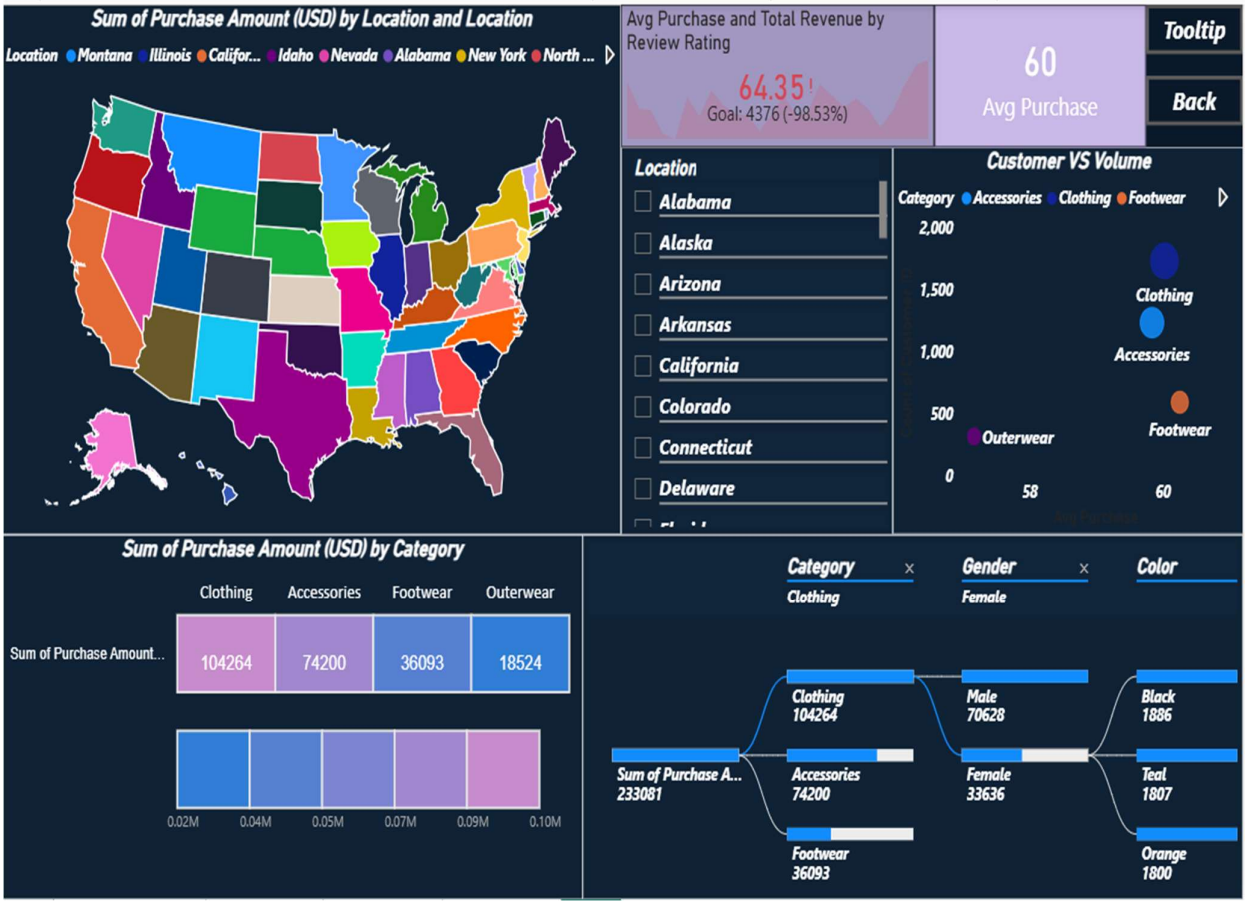

Item Purchased
Search
Shopping Behavior Analysis Dashboard
Next
27K
Sum of Purchase Amount (USD)
458
Count of Customer ID
3.70
Average of Review Rating
11K
Sum of Previous Purchases
58.93
Average of Purchase Amount (...
Sum of Purchase Amount (USD) by Season
7534.00
6981.00
6240.00
6233.00
Winter
Fall
Summer
Spring
Sum of Purchase Amount (USD) by Gender
27K
Male
Gender
Age Group
Adult
Senior
Young Adult
Middle Age
Teen
Discount Applied
No
Yes
Sum of Purchase Amount (USD) by Category
Category
Clothing
Accessories
Footwear
Outerwear
3.9K (14%)
2.1K (8%)
11.9K (44%)
9.2K (34%)
Sum of Purchase Amount (USD) by Item Purchased
Pants
Backpack
Sweater
Jewelry
Hat
Belt
Dress
1695
1537
1409
1347
1345
1302
1252

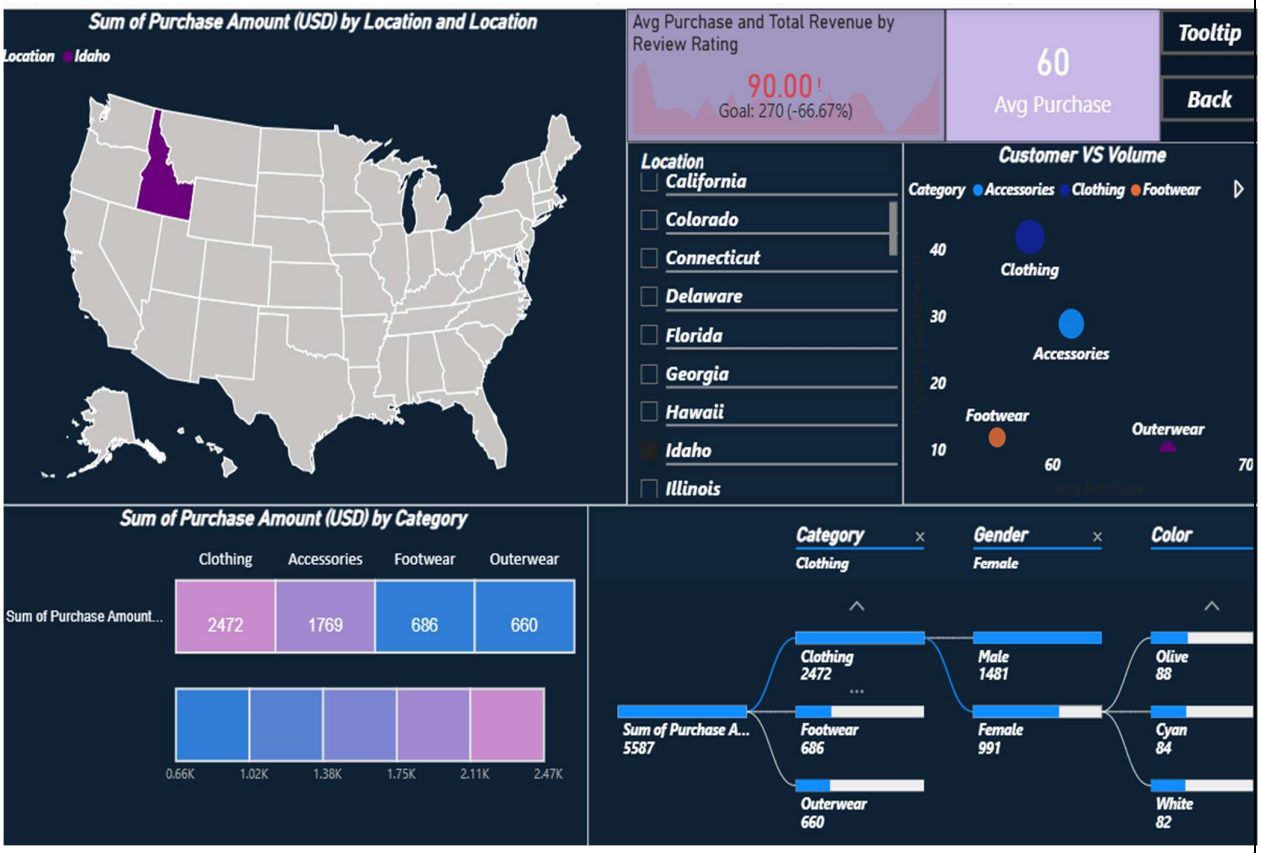
PAGE 2:


Next
Customers
Male %
68.00%
Female %
32.00%
Revenue Contribution by Age Group
68K
65K
48K
43K
9K
Middle Age
Adult
Young Adult
Senior
Teen
Sum of Purchase Amount (USD), Count of Avg Spend and Sum of Avg Spend
233.08K
233K
4K
466K
How Spending Changes with Age
Sum of Review Rating by Shipping Type
100%
Free Shipping
2.51K
Standard
99.49%
Express
97.22%
Next Day Air
96.08%
Store Pickup
96%
2-Day Shipping
93.97%
94%
Count of Customer ID by Category and Discount Applied
Increase
Decrease
Total
Other
Item Purchased
Backpack
Belt
Blouse
Boots
Coat
Dress
Gloves
Handbag
Total Revenue by Subscription Status


Next
Customers
Male %
74.13%
Female %
25.87%
Revenue Contribution by Age Group
4K
2K
0K
3.3K
1.9K
1.6K
1.6K
0.3K
Adult
Middle Age
Senior
Young Adult
Teen
Sum of Purchase Amount (USD), Count of Avg Spend and Sum of Avg Spend
8.55K
0K
8636
17K
How Spending Changes with Age
100
80
60
40
20
94.00
30.00
90.00
85.00
37.00
44.00
79.75
88.67
26.00
94.00
75.67
20
30
40
50
60
70
Sum of Review Rating by Shipping Type
100%
Store Pickup
101.50
2-Day Shipping
98.72%
Next Day Air
95.57%
Express
94.88%
Standard
83.45%
Free Shipping
56.06%
56.1%
Count of Customer ID by Category and Discount Applied
Increase
Decrease
Total
Other
150
100
143
Accessories
Item Purchased
Backpack
Belt
Blouse
Boots
Coat
Dress
Gloves
Handbag
Total Revenue by Subscription Status
6K
4K
2K
0K
No
Yes

Page 3


Sum of Purchase Amount (USD) by Location and Location
Location
Montana
Illinois
Califor...
Idaho
Nevada
Alabama
New York
North ...
Avg Purchase and Total Revenue by Review Rating
64.35
Goal: 4376 (-98.53%)
60
Avg Purchase
Tooltip
Back
Location
Alabama
Alaska
Arizona
Arkansas
California
Colorado
Connecticut
Delaware
Customer VS Volume
Category
Accessories
Clothing
Footwear
2,000
1,500
1,000
500
0
Clothing
Accessories
Footwear
Outerwear
58
60
Sum of Purchase Amount (USD) by Category
Clothing
Accessories
Footwear
Outerwear
Sum of Purchase Amount...
104264
74200
36093
18524
0.02M
0.04M
0.05M
0.07M
0.09M
0.10M
Category
Clothing
Gender
Female
Color
Sum of Purchase A...
233081
Clothing
104264
Accessories
74200
Footwear
36093
Male
70628
Female
33636
Black
1886
Teal
1807
Orange
1800



PROBLEM STATEMENTS:

- Retail organizations generate large volumes of customer shopping data across transactions, product categories, and customer segments, which remains underutilized without proper analysis.
- Decision-makers face difficulties in understanding customer purchasing behavior and identifying high-performing products and customer segments.
- The reliance on static reports and raw datasets makes it challenging to track sales trends and compare performance across categories efficiently.

- Evaluating the impact of discounts, pricing strategies, and promotional activities is complex without interactive analytical tools.
- There is a need for an interactive Power BI dashboard that provides real-time insights, advanced visualizations, and dynamic exploration to support data-driven business decisions.

FINDINGS:

1. Customer Purchase Patterns

- A significant portion of customers make repeat purchases, indicating strong customer retention.
- Shopping frequency varies across customer segments, with a small group contributing to a higher share of total sales.

2.Product Category Performance

- Certain product categories generate higher revenue and purchase volume, making them key drivers of overall sales.
- Some categories show lower demand, highlighting opportunities for promotions or product optimization.

3.Spending Behavior

- Customers with higher purchase frequency tend to have higher average spending.
- A clear variation is observed in spending patterns across different customer groups.

4.Impact of Discounts and Pricing

- Discounts positively influence purchase decisions, leading to increased sales volume.
- However, excessive discounting reduces profit margins, indicating the need for balanced pricing strategies.

5.Customer Segmentation Insights

- Specific customer segments contribute disproportionately to total revenue.

- Identifying and targeting high-value customers can significantly improve business performance.

6.Sales Trends

- Seasonal and periodic trends are observed in customer purchases.
- Peak shopping periods show increased transaction volume and higher revenue generation.

7.Customer Satisfaction Indicators

- Higher spending customers generally show better satisfaction levels.
- Customer engagement and satisfaction play a crucial role in repeat purchases**.**

Conclusion

The Customer Shopping Behavior Dashboard successfully transforms raw retail data into meaningful and actionable insights through interactive visualizations and key performance indicators. By consolidating customer demographics, purchasing patterns, product category performance, and discount impact into a single analytical platform, the dashboard enables stakeholders to gain a comprehensive understanding of customer behavior.

The analysis highlights the importance of identifying high-value customers, optimizing product categories, and applying balanced pricing and discount strategies to improve overall business performance. Interactive filtering and advanced Power BI visuals allow decision-makers to explore trends dynamically, uncover hidden patterns, and respond proactively to changing customer preferences.